

NEWS RELEASE 1-JUN-2026

World record: 450 Tb/s transmission over a metropolitan link using legacy optical fiber

Reports and Proceedings

NATIONAL INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY (NICT)

Highlights

- A new world record for optical data transmission: 450 Tb/s over a metropolitan network.
- The transmitted signal occupied 42.4 THz of bandwidth - the widest ever sent through an optical fiber.
- The result was achieved over already-installed (legacy) optical fiber in London, UK, linking University College London to the Telehouse North data center.

Abstract

The National Institute of Information and Communications Technology (NICT, President: OHNO Hideo, Ph.D.), together with 5 international research partners, has demonstrated a record-breaking 450 terabits per second (Tb/s) optical transmission - the first time such a rate has been achieved over a field-deployed legacy fiber. The link, in London, UK, connects University College London (UCL) to the Telehouse North data center.

This work is part of a long-standing collaboration between NICT and UCL.

The team introduced new optical-amplifier technologies that support ultra-wideband signals. Conventional commercial systems use up to about 10 THz of bandwidth, covering the C- and L-bands. The new system uses the O-, E-, S-, C- and L-bands together, more than quadrupling the available bandwidth to a record 42.4 THz. The resulting data rate of 450 Tb/s surpasses the previous records of 402 Tb/s and 430 Tb/s, set in 2024 and 2025 over laboratory fibers. Unlike those earlier demonstrations, the new experiment used real, already-installed fibers from the UK National Dark Fibre Facility (NDFF). This is therefore the closest demonstration to date of how the full capacity of existing fiber infrastructure could be unlocked, paving the way for the next generation of networks needed to support AI services and beyond-5G mobile systems.

This achievement was reported as a postdeadline paper at the 2026 Optical Fiber Communication Conference (OFC2026) on Thursday, March 19, 2026, at the Los Angeles Convention Center, Los Angeles, California, USA.

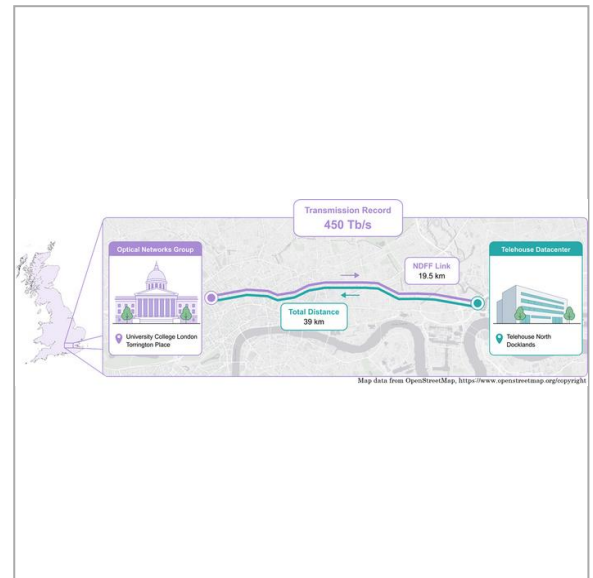


IMAGE:

FIGURE 1. SIMPLIFIED DIAGRAM OF THE TRANSMISSION EXPERIMENT

[view more >](#)

CREDIT: NATIONAL INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY (NICT) MAP DATA FROM OPENSTREETMAP, [HTTPS://WWW.OPENSTREETMAP.ORG/COPYRIGHT](https://www.openstreetmap.org/copyright)

Background

New services such as AI, self-driving vehicles and beyond-5G mobile communications are placing growing demands on telecommunication networks. To meet these needs, the technique known as multi-band wavelength-division multiplexing (WDM) has been developed to make use of the full capacity of today's optical fibers. In earlier work, we demonstrated transmission across the O-, E-, S-, C-, L- and U-bands, which together cover most of the low-loss region of an optical fiber. That demonstration, however, was carried out only under ideal laboratory conditions.

In this new experiment, we exceeded the previous bandwidth and data-rate limits using new amplifier technologies covering the O-, E-, S-, C- and L-bands. Crucially, the experiment was performed over a field-deployed fiber link, which has significantly higher loss than a laboratory fiber due to splices, connectors and previous fiber-cut repairs. The link, installed in London, UK, runs from the UCL campus to the Telehouse North data center as shown in Figure 1.

Achievements

Working with its partners, NICT developed an optical-fiber transmission system with the widest bandwidth ever reported, 42.4 THz, and used it to send a signal through a field-deployed fiber in London, UK. With a total throughput of 450 Tb/s, the signal sets a new capacity record for a standard optical fiber. The wideband WDM signal contained up to 1,273 individual wavelength channels across the O-, E-, S-, C- and L-bands, spanning 42.4 THz of bandwidth (1,264.0 nm to 1,617.8 nm) as shown in Table 1. The signal was sent over 39 km of fiber, most of it running underground, between the UCL campus and the Telehouse data center in London Docklands, and back again. High data rates were obtained using dual-polarization quadrature amplitude modulation (DP-QAM) with up to 256 symbols per constellation. The generalized mutual information (GMI)-based estimated data rate after 39 km transmission reached 450 Tb/s. This surpasses the previously reported highest data rate in single-mode fiber (SMF). Table 1 compares this result with our results of past wideband transmission experiments. These results show that multi-band wavelength-division multiplexing can unlock previously untapped capacity in standard optical fibers.

New ultra-high-capacity optical-fiber technologies will be essential for communications beyond 5G. To keep deployment costs and timelines manageable, they need to be compatible with the fiber infrastructure that is already in the ground. Building on existing networks in this way enables faster rollout, better use of resources and reliable high-speed connectivity for tomorrow's digital services. The paper containing these results was presented at the postdeadline session of the 2026 Optical Fiber Communication Conference (OFC2026) on Thursday, March 19, 2026, at the Los Angeles Convention Center, Los Angeles, California, USA.

Future Prospects

NICT will continue to drive research and development on new technologies, components and fibers that open up additional transmission windows, for both near- and long-term applications. NICT also aims to extend the reach of these wideband, ultra-high-capacity systems and to broaden their compatibility with

field-deployed fibers.

Previous NICT Press Releases

- Novel Transmission Technique Enables World Record 430 Tb/s in a Commercially Available, International-Standard-Compliant Optical Fiber

<https://www.nict.go.jp/en/press/2025/11/11-1.html>

- World Record 402 Tb/s Transmission in a Standard Commercially Available Optical Fiber

<https://www.nict.go.jp/en/press/2024/06/26-1.html>

- World Record 301 Tb/s Transmission in a Standard Commercially Available Optical Fiber

<https://www.nict.go.jp/en/press/2024/01/29-1.html>

- World's First Successful Transmission of 1 Petabit per Second in a Standard Cladding Diameter Multi-core Fiber

<https://www.nict.go.jp/en/press/2022/05/30-1.html>

Reference

Optical Fiber Communication Conference (OFC) 2026, Postdeadline Session

Title: 450 Tb/s GMI, 42.4 THz, OESCL-Band Transmission Over a Field-Deployed Fiber

Authors: Ruben S. Luis, Jiaqian Yang, Romulo Aparecido, Mindaugas Jarmolovicius, Eric Sillekens, Ronit Sohanpal, Zelin Gan, Aleksandr Donodin, Vitaly Mikhailov, Jiawei Luo, David DiGiovanni, Nicolas Fontaine, Lauren Dallachiesa, Mikael Mazur, Roland Ryf, Haoshuo Chen, David Neilson, Ian Phillips, Wladek Forysiak, Sergei Turitsyn, Daniele Orsuti, Hideaki Furukawa, Robert Killey, and Polina Bayvel.

METHOD OF RESEARCH

Experimental study

SUBJECT OF RESEARCH

Not applicable

ARTICLE TITLE

450 Tb/s GMI, 42.4 THz, OESCL-Band Transmission Over a Field-Deployed Fiber

ARTICLE PUBLICATION DATE

19-Mar-2026

Disclaimer: AAAS and EurekAlert! are not responsible for the accuracy of news releases posted to EurekAlert! by contributing institutions or for the use of any information through the EurekAlert system.

Media Contact

Norio KOBAYASHI
National Institute of Information and Communications Technology (NICT)
publicity@nict.go.jp
